Hugo Marquerie 12/01/2026

Teorema de diferenciación de Lebesgue

Teorema 1 (de diferenciación de Lebesgue). *Sea* $f \in \mathcal{L}^1_{loc}(\mathbb{R})$, *entonces*

$$\lim_{h \to 0} \frac{1}{2h} \int_{x-h}^{x+h} f(t)\,\mathrm{d}t = f(x) \quad \textit{c.t.p. } x \in \mathbb{R}.$$

***Demostración*:** Por el `lem-diferenciacion-lebesgue-l1-imp-loc-l1/`**??**, basta probarlo para $f \in \mathcal{L}^1(\mathbb{R})$, que es el caso del `teo-diferenciacion-lebesgue-l1/`**??**, luego queda demostrado directamente. ■

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